c:\hpc-23-feb-2026\.\29\client.c

```
#include<stdio.h>
#include<sys/socket.h>
#include<string.h>
#include<netinet/in.h>
#include<arpa/inet.h>
#include<unistd.h>
int main()
{
int file_descriptor;
struct sockaddr_in server_address;
char buffer[1025];
int n;
file_descriptor=socket(AF_INET,SOCK_STREAM,0);
if(file_descriptor<0)
{
printf("Unable to create socket\n");
return 1;
}
// initialize all the bytes of server_address structure to zero
bzero(&server_address,sizeof(server_address));
server_address.sin_family=AF_INET;
server_address.sin_port=htons(3500);  // server port
if(inet_pton(AF_INET,"127.0.0.1",&server_address.sin_addr)<0)
{
printf("Unable to setup server ip\n");
return 1;
}
if(connect(file_descriptor,(struct sockaddr *)&server_address,sizeof(server_address))<0)
{
printf("Unable to connect to server\n");
return 1;
}
n=read(file_descriptor,buffer,1024);
if(n==0)
{
printf("Server is not saying anything\n");
}
else
{
buffer[n]='\0';
printf("Server said [%s]\n",buffer);
}
return 0;
}
```

c:\hpc-23-feb-2026\.\29\server.c

```
#include<sys/socket.h>
#include<arpa/inet.h>
#include<netinet/in.h>
#include<stdlib.h>
#include<unistd.h>
#include<string.h>
#include<stdio.h>
int main()
{
int server_file_descriptor;
int client_file_descriptor;
char buffer[1025];
struct sockaddr_in server_sockaddr;
server_file_descriptor=socket(AF_INET,SOCK_STREAM,0);
if(server_file_descriptor<0)
{
printf("Unable to setup socket\n");
return 1;
}
bzero(&server_sockaddr,sizeof(server_sockaddr));
server_sockaddr.sin_port=htons(3500);
server_sockaddr.sin_family=AF_INET;
server_sockaddr.sin_addr.s_addr=htonl(INADDR_ANY);
if(bind(server_file_descriptor,(struct sockaddr *)&server_sockaddr,sizeof(server_sockaddr))<0)
{
printf("Unable to bind socket to port 3500\n");
return 1;
}
printf("INADDR_ANY %d\n",INADDR_ANY);
printf("SOMAXCONN %d\n",SOMAXCONN);
if(listen(server_file_descriptor,SOMAXCONN)<0)
{
printf("Unable to put socket in listening mode\n");
}
while(1)
{
printf("Waiting to accept connection.....\n");
client_file_descriptor=accept(server_file_descriptor,(struct sockaddr *)NULL,NULL);
printf("Connection accepted\n");
// 2nd argument at later stage will be address of structure created from struct sockaddr
// 3rd argument will be address of int
// ideally the following code should go on separate thread
strcpy(buffer,"Welcome to server");
write(client_file_descriptor,buffer,strlen(buffer));
close(client_file_descriptor);
}
return 0;
}
```